04-14-2025

Final: May 1 12-3 BLC A400

Hugget (1993)

$e \begin{cases} e_l \\ e_h \end{cases} \Rightarrow$ endowment stochastic w/ trasition prob $\pi(e'|e)$ and CRRA: $U(c) = \frac{c^{1-\sigma}}{1-\sigma}$

Allows borrowing b/w agents and mkt. clear b/w each other

pay $aq \rightarrow a$ recovered

$a \geq \underline{a}$ is borrowing limit

$$v(a,e) = \max_{c,a} \{u(c) + \beta \sum_{e'} \pi(e'|e) v(a',e')\}$$

s.t.

$$c + aq = a + e$$

$F(a,e)$ CDF, $f(a,e)$ PDF

Stationary C.E.:

i) policy functions $a'(a,e)$, $c(a,e)$ and value $f(\cdot)$ $v(a,e)$,

ii) invariant distribution $F(a,e)$,

iii) price q,

iv) aggregate variables,

such that

i) policy $f(\cdot)$'s + value $f(\cdot)$ solve individual aptimization problem

ii) markets clear:

consumption: $\sum_e \int_{\underline{a}}^{\infty} c(e,a) f(e,a)\, da = \sum_e \int_{\underline{a}}^{\infty} e f(e,a)\, da$

credit: $\sum_e \int_{\underline{a}}^{\infty} a' f(e,a)\, da = 0$

iii) The distribution of agents over states is Stationary

- $F(e', a') = \pi(e' | e_h) F(e_h, a_h) + \pi(e' | e_\ell) F(e_\ell, a_\ell)$

  s.t. $a' = a'(a_h, e_h)$

  $a' = a'(a_\ell, e_\ell)$

  ↳ some transition will be zero

- Alternative:

  $F = H(F)$ (law of motion)

Conditions for Unique Equilibrium:

i) $\underline{a} + e_\ell + \underline{a}q > 0$ (consumption at the bottom > 0)

ii) $\beta < q$ (like $\beta(1+r) < 1$)

↳ precautionary motive to save

iii) $\pi(e_h | e_h) \geq \pi(e_h | e_\ell)$

↳ $e_h = 1$; $e_\ell = 0.1$, $\sigma = 1.5$ in paper

→ Algorithm:

i) guess q

ii) solve consumer optimization problem for a' + c

iii) compute invariant distribution $F(a, e)$

iv) compute $\sum_e \int_{\underline{a}}^{\infty} a'(a,e) f(a,e) da$, compare w/ 0

v) update q depending on how above deviates

vi) repeat ii-v

Result:

consider $\underline{a} = [-2, -4, -8]$; $r = [-7.1\%, 2.3\%, 4\%]$

↳ more borrow ⇒ incentive to save w/ $r \uparrow$

↳ less borrow ⇒ disincentive saving w/ $r \downarrow$

↳ as $\underline{a} \to -\infty$, closer to rep agent model

↳ as borrowing capacity shrinks, heterogeneous agent

Aiyagari (1994):
- wrote paper w/ specific goal: precautionary save contribution to agg K.

ε_t is efficiency following AR(1) s.t. $\varepsilon \in [\varepsilon_{min}, \varepsilon_{max}]$
$c_t + a_{t+1} = w\varepsilon_t + (1+r)a_t$

Natural borrowing limit:

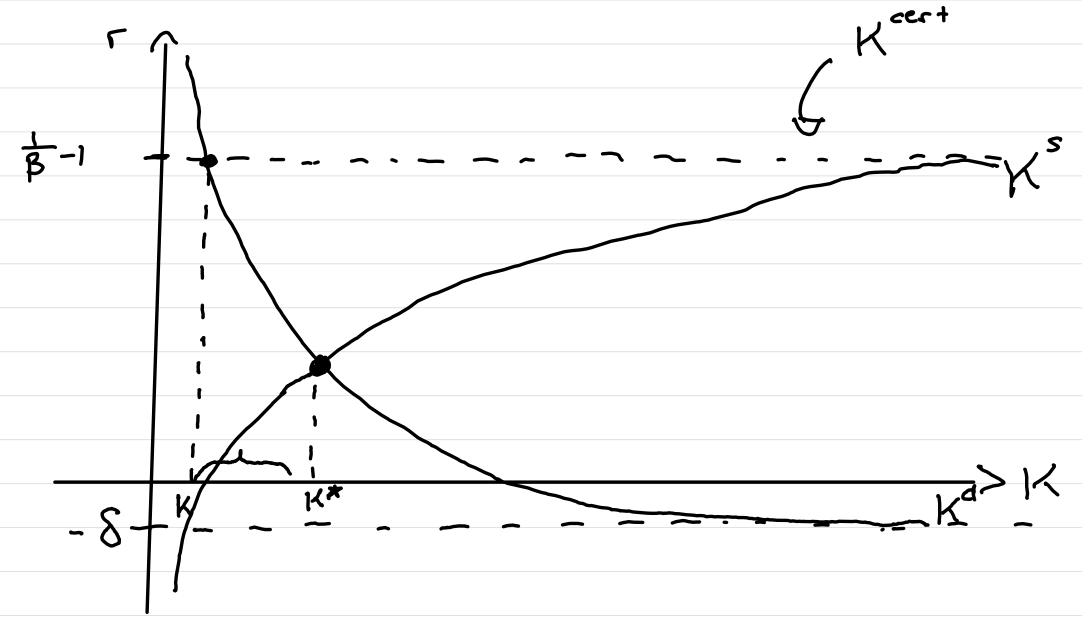
$$\underline{a} = \frac{w \cdot \varepsilon_{min}}{r}$$

$$c = w\varepsilon_{min} + (1+r)a - a$$
$$= w\varepsilon_{min} + ra > 0$$

$$V(a,\varepsilon) = \max_{c,a'} \{u(c) + \beta E[v(a',\varepsilon')]\}$$
s.t.
$c = w\varepsilon_{min} + ra > 0$
$a \geq -b$ s.t. $b :=$ borrowing limit

Firm:
$$\Pi = K^\alpha N^{1-\alpha} - wN - (r+\delta)K \rightarrow \max_{K,N}$$
FOC:
$$\left.\begin{aligned} r+\delta &= \alpha\left(\frac{K}{N}\right)^{\alpha-1} \\ w &= (1-\alpha)\left(\frac{K}{N}\right)^{\alpha} \end{aligned}\right\} K^d = \left(\frac{\alpha}{\delta+r}\right)^{1/1-\alpha} N$$

MCC:
K: $K^s = \int a'(a,\varepsilon)\,dF(a,\varepsilon) = K^d$
N: $N = \int \varepsilon\, dF(a,\varepsilon)$



gap due to precautionary saving

decisions change, but must take eq. K into account.

Calibration:

$U(c) = c^{1-\bar{\sigma}}/1-\bar{\sigma}$ s.t. $\bar{\sigma} = [1, 3, 5]$

$\log \varepsilon = \rho \log_{t-1} + (\sigma^2(1-\rho^2))^{1/2}\gamma_t, \ \gamma_t \sim N(0,1)$

$\sigma = [0.2, 0.4]$

$\rho = [0, 0.3, 0.6, 0.9]$

Algorithm:

i) guess r

ii) find $(\frac{K}{N}) \Rightarrow$ find w (from FOC's) be specific on exam

iii) solve consumer optimization problem

iv) find invariant distribution

v) compute $K = \int a'(a,\varepsilon)\, dF(a,\varepsilon)$

vi) using firm FOC, update $r_{new} = \alpha K^{\alpha-1} N^{1-\alpha} - \delta$

s.t. $N = \int \varepsilon\, dF(a,\varepsilon)$

vii) if same $\Rightarrow$ done

else: $\xi r_{new} + (1-\xi) r = r \Rightarrow$ repeat 2-7

Results:

- Saving rate = investment/output = $\frac{K' - (1-\delta)K}{K^{\alpha} N^{1-\alpha}}$
  $= \delta \left(\frac{K}{N}\right)^{1-\alpha}$ in S.S.
  $\Rightarrow r + \delta = \alpha \left(\frac{K}{N}\right)^{\alpha - 1} \Rightarrow \; = \frac{\delta\alpha}{r+\delta}$

- agg. saving:
  - no uncty: 23.67% $\quad (\beta(1+r^{rep}) = 1)$
  - w/ uncty: 24.14%
    $\bar{\sigma} = 1, \rho = 0, \sigma = 0.2$
  - w/ uncty: 37.63
    $\bar{\sigma} = 5, \rho = 0.9, \sigma = 0.4$

  ↳ precautionary is major contributor to agg. K

All 3 SIM models are building block of modern macro

Very useful and powerful

Limitations:

- wealth inequality

Earnings Inequality	Wealth Inequality	
3.6	-0.9	Q_1
9.9	0.8	Q_2
15.3	4.4	Q_3
22.7	13.0	Q_4
48.5	82.7	Q_5
10.9	13.7	90-95
13.1	22.8 } > 50%	95-99
8.0	30.9 }	Top .01